Smart City Traffic Final Capstone Report

Data analytics Python engineering and machine learning

This report brings together the methodology, findings and limitations from all three parts of the Smart City Traffic Capstone. The project uses historical traffic and weather observations from one corridor to develop an analytics workflow, predictive models and a local decision-support prototype.

The main findings are that traffic varies strongly with time of day, the temperature–traffic relationship is weak, and the saved Random Forest predicts held-out traffic volume with MAE 250.84 vehicles per hour and R-squared 0.9448. Recommendations rank supplied travel times; they do not provide live traffic, alternative routes or journey times. No real accident dataset was supplied or sourced: classification uses a documented proxy.

Part	Scope	Evidence
1	SQL statistics probability and Power BI	48,204 original records and 40,575 distinct recorded hours
2	Cleaning features charts logging and CLI	48,187 cleaned rows and 30 feature columns
3	Models SHAP tracking recommendations API and responsible AI	Seven task notebooks and chronological held-out evaluation

Report organisation

Each part below retains its reviewed methodology and findings. Original Part 1 statistics, cleaned Part 2 summaries and Part 3 held-out model evaluations describe different stages and should not be treated as interchangeable results. Detailed commands and execution evidence are provided in the repository READMEs and execution output documents.

Part 1 Data Analytics

This report explains the main findings from the four tasks. We have used SQL queries, Excel and Power BI to study the traffic patterns. The main aim of this project is to help the smart-city mobility team understand the busy periods and make better planning decisions.

The dataset contains 48204 records and 40575 different recorded hours. Some hours appear more than once. I have used all records, while the total hours card counts each recorded hour once.

Task 1 Yearly traffic and holidays

Recorded traffic totals fell in 2014 and 2015, then increased in 2016 and 2017. The largest decrease was 44.17% between 2013 and 2014. The highest recorded total was in 2017, at about 35.43 million. This was 20.12% higher than 2016.

However, all the years do not have the same amount of data. For example, 2012 starts in October, and there are missing periods in other years. Repeated hours also affect the totals. The team should check these gaps before using the results to say that actual traffic increased or decreased.

The holiday results did not show a clear link between temperature and traffic. The record labelled New Year's Day was 4.68C warmer in 2017 than in 2016, but traffic fell from 1513 to 798. The 2017 record is dated 2 January, while the 2016 record is dated 1 January. Labor Day also showed no consistent pattern. These results only cover midnight records, so they cannot describe the whole holiday. No New Year's Day record was availble for 2015.

Task 2 Traffic statistics and temperature

The average traffic volume was 3259.82, and the median was 3380. The population standard deviation was 1986.84, and the population variance was 3947533.43. Traffic values ranged from 0 to 7280, giving us a range of 7280.

These results show that traffic changes a lot between records. The average alone does not show how busy or quiet a particular hour can b e. The team should study these periods seperately.

The correlation between temperature and traffic was 0.1303. This shows a weak positive link. Warmer temperatures sometimes came with more traffic, but the relationship was small. This does not prove that temperature causes traffic to increase. Time of day and people's travel habits might also explain the pattern.

Task 3 Probability and congestion

Congestion was defined as traffic volume above 5500. Using this rule:

14.73% of records had congestion.

27.78% had clear weather.

3.66% had both congestion and clear weather.

Among congestion records, 24.83% had clear wether and 26.30% had temperatures above 292K.

The independence calculations did not give equal results. This indicates that the events were not exactly independent in this dataset.

The odds ratio for congestion in clear weather compared with cloudy weather was 0.7354. Congestion had lower odds in clear weather. However, this does not prove that weather caused the difference.

Traffic volume aso does not tell us how slowly vehicles were moving.

Task 4 Power BI dashboard

The original data had 9 columns and no blank or null values. We checked the data types and added Hour, Temperature in Celsius, Traffic Category and Datecolumns.

The daily chart shows frequent changes in traffic. In 2017, the highest hourly average was at 4 PM, at about 5798. The lowest was at 3 am, around 382.

Clouds had the highest average traffic at 3618.45. Squall had the lowest at 2061.75. The difference was 1556.70.

The scatter plot showed no clear temperature–traffic pattern. We havce found 10 invalid(outliers)temperature readings at −273.15°C and hid them in the second plot.

The cards showed 40575 recorded hours, average traffic of 3259.82 and average temperature of 8.06C. The temperature average still includes the invalid readings. All three filters worked correctly in our combined test.

Summary

The team should focus monitoring on busy afternoon hours and consider quieter hours for road maintenance. It should also check missing hours, , invalid temperatures , repeated records and zero traffic values before further analysis. Weather might help with planning, but it should be considered along with time of day and other travel information.

Part 2 Python Traffic Analytics

I have used Python to clean the traffic data, add useful features and create three charts. I also made a small command-line application. The results show that average traffic is higher during the day and on weekdays.

Task 1: Data Pipeline Construction

The original CSV had 48204 rows and 9 columns. I checked that the required columns were present before cleaning the data. I made category text consistent, converted dates and numbers, and checked for missing or invalid values.

I removed 17 exact duplicate rows. I replaced 10 temperature readings of 0 Kelvin and one rainfall reading above 9000 mm. For each replacement, I used the median of valid readings from the same calendar month across all years. This left 48187 rows and 9 columns in traffic_cleaned.csv.

Each cleaning step writes its own log message. Warning messages s how how many rows changed and why. The script checks the cleaned data before saving it.

Task 2: Feature Engineering

I added the hour, day of week and weekend indicator. I used sine and cosine to encode the hour so that 23:00 and 00:00 are close in the cycle. I also added weather-category columns and rain and snow indicators. I scaled temperature and rainfall to values between 0 and 1. I used the 25th and 75th percentiles of traffic volume to create three traffic categories. Low is at or below 1192.5, Medium is above 1192.5 and at or below 4933, and High is above 4933.

These categories describe lower and higher traffic volume in this dataset. They do not measure road congestion directly. The saved feature dataset has 48187 rows and 30 columns.

Task 3: Visualise Traffic Patterns

As instructed, I saved three Matplotlib charts: traffic by hour, weekday and weekend traffic, and traffic by weather category.

The hourly chart shows the lowest average traffic at 03:00, with 371.09 vehicles. The highest average is at 16:00, with 5663.76 vehicles. Traffic rises quickly in the morning and stays higher during the day. Weekday average traffic is 3533.29, compared with 2570.70 on weekends. The difference is 962.59. This may reflect work and school travel, but the chart alone cannot show the reason.

Clouds has the highest weather-category average at 3617.99. Squall has the lowest at 2061.75, but there are only 4 Squall records. Smoke also has few records, with 20 observations. Small groups need careful interpretation.

Task 4: Build a Mini Traffic Analytics Application

The application supports three commands. Users can query a specific date and time, find traffic above a chosen threshold, or compare weekdays and weekends. The chosen threshold is separate from the traffic-category percentiles.

I thave ested all three commands. The date query returned traffic volume 5545 for 2012-10-02 at 09:00:00. The high-traffic query listed records above 7000. The day comparison matched the chart averages. Invalid dates and non-numeric thresholds gave clear ERROR messages without a traceback and returned exit code 1.

Task 5: GitHub and Reproducibility

I saved the work in separate Git commits for each completed coding task. The README explains the folder structure, required packages, script order and application commands. It also explains logging to the console and pipeline.log. INFO shows normal steps, WARNING shows recoverable data issues, ERROR shows failures, and DEBUG shows internal calculations when enabled.

Limits of the findings

The averages use all cleaned records. Some timestamps have more than one weather record, so the averages are based on records rather than unique hours. There are also gaps in the original data. Weather averages may reflect differences in time of day or season. These results show patterns, but do not prove that weather causes congestion.

Part 3 Machine Learning and AI

Purpose and main findings

I Have developed an intelligent mobility solution using the Metro Interstate Traffic Volume dataset and the preparation workflow from Part 2. This report covers the methodology and findings across all seven Part 3 tasks. The solution predicts traffic volume, classifies a documented accident-risk proxy, explores traffic patterns, recommendstravel times and demonstrates experiment tracking, deployment and monitoring.

Random Forest achieved a held-out traffic-volume MAE of 250.84 vehicles per hour and R-squared of 0.9448. The neural network achieved MAE of 256.94 and R-squared of 0.9455. I have used the saved Random Forest for the recommendation engine and deployment simulation. Its lower test MAE supports this choice, although it does not establish that it will perform best in every future period. The results also show weaker performance in some conditions and limited severe-weather coverage.

The dataset represents one corridor and historical observations. Recommendations compare supplied travel times, they do not provide alternative routes, live traffic or journey times. No real accident dataset was supplied or sourced.

Data preparation and evaluation design

The original dataset contained 48204 records. The Part 2 preparation workflow corrected text formatting, checked types and missing values, and removed 17 exact duplicate rows, leaving 48187 records. Different weather records can share an hourly timestamp, so I retained these records and kept each timestamp entirely within one split.

For supervised modelling, I treated 10 nonpositive temperature readings and one exceptionally large rain reading above 9000 as missing. Monthly medians fitted on the training data supplied these values. Preparation settings and fitted transformations are saved with the models so that later prediction used the same process.

I split the unique timestamps chronologically into training, validation and test periods rather than randomly mixing future and past observations. The resulting row counts are 35873 for training, 7370 for validation and 4944 for testing. Training has ended on 2 August 2017 at 23:00. Validation ran from 3 August 2017 at 00:00 to 14 April 2018 at 15:00. Testing ran from 14 April 2018 at 16:00 to 30 September 2018 at 23:00.

The common feature set includes temperature, hour rain, snow, cloud cover, day of week, month, weekend and holiday flags, cyclical sine and cosine encodings for hour and day of week, along with rain, snow and low-visibility indicators. Weather categories are one-hot encoded. The 17 input features became 27 transformed features. Scaling and encoding were fitted on training data.

Task 1 Supervised Machine Learning Models

Methodology

I compared Linear Regression with Random Forest for traffic-volume regression and Logistic Regression with Random Forest for classification. Both problems used the common engineered feature set. The Random Forest regressor used 100 trees, maximum depth 20, minimum leaf size 2 and random seed 42. Saved preparation settings and model files support repeatable inference.

The accident-risk proxy combined congestion and weather. Training traffic-volume quartiles are defined as Low, Medium, High and Severe congestion. High or Severe congestion together with severe or low-visibility weather has produced the positive class. The thresholds have been fitted on training data and then applied to validation and test data. This avoids fitting the label thresholds on the future test period.

Regression findings

Model	Validation MAE	Validation R-squared
Linear Regression	826.19	0.7158
Random Forest	317.57	0.9256

Random Forest substantially improved validation MAE and R-squared over the linear baseline. Its held-out test MAE was 250.84 and R-squared was 0.9448. The improvement suggests that nonlinear relationships between time, weather and traffic matter for this dataset. It does not show that these features cause traffic changes.

Classification findings

Model and period	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression validation	0.9507	0.7645	0.9842	0.8605	0.9895
Random Forest validation	0.9769	0.9047	0.9508	0.9272	0.9932
Random Forest test	0.9875	0.9091	0.9664	0.9369	0.9982

Random Forest had the stronger validation F1 score and precision, while Logistic Regression had higher recall. The positive class contained 476 test records. Weather inputs also helped define the target, so these strong scores mainly demonstrate reproduction of the proxy rule. They are not evidence of accurate prediction of observed accidents or injuries.

Task 2 Unsupervised Machine Learning

K means clustering

I have clustered historical traffic conditions using scaled hour sine and cosine, weather severity and traffic volume. I have compared 2 to 6 clusters using silhouette scores with a reproducible sample of 3000 records and selected five clusters, which had the highest tested score of approximately 0.3704. This was an exploratory analysis of the full cleaned historical dataset, separate from supervised test evaluation.

Cluster	Records	Mean traffic volume	Most common hour	Most common weather
0	8043	1079.69	04:00	Mist
1	12787	5080.17	14:00	Clouds
2	9624	3189.59	18:00	Clouds
3	9039	4924.87	09:00	Rain
4	8694	944.87	00:00	Clear

Clusters 0 and 4 represented lower-volume conditions, with different weather patterns. Cluster 1 represented busy conditions with low encoded weather severity. Cluster 3 combined high volume with more severe weather, while cluster 2 represented more moderate traffic. Cluster assignments describe similarity under the chosen features and severity encoding, rather than fixed operationa traffic classes.

Association rules

I discretised time of day, weekday type, weather and congestion. Congestion quartiles for this exploratory analysis came from the full cleaned dataset. I havemined context combinations leading to congestion categories with minimum support 0.01, confidence 0.60 and lift above 1. 38 rules met these criteria.

The highest-lift rule was Weekend, Night and Mist leading to Low congestion. It applied to 558 antecedent records, of which 511 had Low congestion. Confidence was approximately 91.58 %, support 1.06 %and lift 3.663. Low congestion was therefore more frequent within this context than its overall frequency. The rule describes historical association and does not establish that mist reduces traffic.

Task 3 Deep Learning with Explainability

Neural network

I have trained a neural network with hidden layers of 64 and 32 units using ReLU activation and a linear output. Inputs used the same fitted feature preparation as the Random Forest, and the target was standardised using the training statistics. I used the Adam optimiser to adjust the network's weights during training, with a learning rate of 0.001 to control the size of each update.. Early stopping monitored validation loss with patience 10 and restored the best weights. Training finished after 37 epochs, with the best epoch at 27.

Model	Validation MAE	Validation R-squared	Test MAE	Test R-squared
Random Forest	317.57	0.9256	250.84	0.9448
Neural Network	316.44	0.9307	256.94	0.9455

The neural network slightly improved validation results and held-out R-squared, while Random Forest had lower held-out MAE. The small differences do not support an unconditional claim that one model is superior.

Explainability

I have used Tree SHAP on the comparable Random Forest trained for the same traffic-volume problem. This provides an explanation of the Random Forest, not a direct explanation of the neural network. I have 00selected 200 held-out records with a fixed random seed and checked that the base value plus contributions reconstructed predictions.

Hour cosine had the largest mean absolute contribution, approximately 1438.38 vehicles per hour, followed by hour at 447.40, hour sine at 204.08 and day of week at 180.97. These findings show that time patterns strongly influenced this model.

For an example at 15:00 on 18 April 2018, actual traffic volume was 5835 and the prediction was 5801.59. The SHAP base value was 3244.99 and contributions summed to 2556.60. The saved global chart, local waterfall and contribution tables show both overall and individual prediction explanations.

Task 4 Apply an Advanced AI Technique

I selected MLflow because experiment tracking connects model comparison with the later deployment simulation. I have recorded the existing Random Forest and neural network evaluations in the Smart City Traffic Regression experiment, using a local SQLite tracking database and artifact folders.

The two completed runs recorded parameters, validation and test MAE and R-squared, model version v1, and relevant model, preparation and evaluation artifacts. The neural network run also retained training evidence. I vhave erified the recorded metrics and artifact inventory against the saved results. These runs document existing evaluations, they do not represent additional model training.

MLflow adds traceability by keeping model settings, results and artifacts together. It supports review and reproduction of the selected model. This implementation uses local tracking and copied artifacts rather than a managed model registry or an automated promotion process. Artifact copies increases storage, and relative artifact locations require the working directory when using the local tracking interface.

Task 5 Build a Traffic Recommendation System

I have built a reusable recommendation module that validates the candidate travel records, applies the saved preparation workflow and predicts traffic volume with the saved Random Forest. Inputs include date and time, weather measurements, weather category and holiday status. Calendar and cyclical features are derived from the supplied timestamp.

The engine ranks candidate hours by predicted traffic volume, using time to resolve ties(proprtising the earlier timestamps), and generates a plain-language recommendation. It preserves the relationship between each candidate and its prediction. Validation checks covered shuffled input order, duplicate timestamps, unknown weather and invalid cloud cover.

For five supplied clear-weather weekday candidates on 7 May 2018, the predictions were:

Candidate hour	Predicted vehicles per hour
06:00	5692.33
07:00	5475.10
08:00	5871.44
09:00	4956.37
10:00	4204.31

The engine recommended 10:00 among these supplied choices. Its predicted volume was approximately 1667 vehicles per hour below the highest-volume candidate at 08:00. This example demonstrates thetiming advice for one corridor. It does not guarantee a shorter journey or account for current incidents, and the candidate weather values must come from the caller.

Task 6 MLOps and Deployment Simulation

Versioning and deployment

The deployment manifest records the selected model version, artifact locations, SHA-256 hashes and evaluation metrics. The MLflow comparison retains the two evaluated model families and their v1 results. No second trained release or automatic retraining process iscreated.

I have implemented a FastAPI application with GET /health and POST /predict. The prediction endpoint validates the records, applies the reusable preparation function and returns model version, units, timestamps and predicted traffic volumes in input order. Model and preparation hashes are verified when

the saved application is loaded.

Tests checked the saved application against thenotebook inference, using exact comparisons for metadata and timestamps and a tight numerical tolerance for predictions. Both health and valid prediction requests returned HTTP 200. Invalid inputs are rejected with HTTP 422. Live local HTTP tests and browser tests have confirmed the running application, including rejection of cloud cover above 100.

Monitoring and alerting

The monitoring function compares batch means for temperature, rain, cloud cover and hour with training scaler statistics. A batch requires at least 100 records. An absolute mean shift greater than one training standard deviation triggers ALERT, otherwise the selected checks return PASS. This is a Python function in the saved application rather than an HTTP endpoint.

A synthetic batch of 240 reference-like records returned PASS. A temperature-shifted batch of 240 records, with temperature increased by two training standard deviations, returned ALERT. A five-record batch is rejected with the expected minimum-size error. These checks demonstrate monitoring behaviour, rather than prove that the threshold is suitable for operations.

Mean checks can miss changes in variance, tails or other relationships. PASS does not guarantee accurate predictions. Monitoring prediction accuracy requires observed traffic volumes. Operational use would require threshold calibration, investigation of alerts and validation before replacing models.

Task 7 Responsible and Sustainable AI

Coverage and uneven errors

I have analysed the saved 4944 held-out predictions without retraning either model. I have alsoe checked timestamp and actual-volume alignment before attaching weather labels. This avoided a timestamp-only join that could duplicate records sharing an hour.

Random Forest MAE is ranged from 97.14 at night to 330.02 in the morning. Neural network MAE ranged from 116.40 at night to 324.96 in the morning. Different traffic levels affect the interpretation of absolute error, so these differences alone do not establish unfairness.

Model	Weekday MAE	Weekend MAE
Random Forest	222.63	320.35
Neural Network	234.90	311.25

Both the models had higher weekend MAE despite lower average weekend volume. This warrants an investigation, but the analysis does not identify a cause. Weekdays contained 3517 records and weekends 1427.

Snow had the highest observed weather-group MAE: 1125.13 for Random Forest and 1164.69 for the neural network. Both overpredicted by approximately 1053 vehicles per hour on average, but only 47 snow records supported this result. Fog had 112 records and MAE of 444.55 and 435.23 respectively. Smoke had only two records and squall had none in the test period.

Groups below 100 records have been flagged as limited evidence. This is a reporting rule, not a statistical guarantee. The April to September test period does not cover a complete annual weather cycle. Repeated hourly timestamps also mean that the records are not all independent.

Proxy risks and governance

The accident-risk proxy is based on the selected congestion and weather conditions rather than observed accidents. Strong classification results must not be presented as validated accident prediction. It should not be used to declare journeys as safe, impose penalties or guide emergency decisions.

Recommendations should also support human judgement alongside current road conditions, incidents and official information. Model versions, preparation settings, metrics and hashes should be retained. Proposed replacement models require validation and group error review. Monitoring alerts should in-turnprompt investigation of data quality, seasonal changes and performance rather than automatic retraining. More representative severe-weather data and evaluation across additional corridors and periods are needed before wider use.

Sustainability

I have reused saved models for recommendation and deployment. Early stopping limited neural network training to 37 epochs rather than the maximum 100. Energy and emissions were not measured, so resource savings cannot be quantified and neither model is claimed to be greener.

MLflow copies improve traceability while increasing storage. Retention policies should preserve the required evidence and archive obsolete versions. Future comparisons should measure training time, inference latency, memory use and, where feasible, energy consumption. The separate responsible and sustainable AI report provides the bias, fairness, governance and sustainability discussion with supporting group result files.

Conclusion and supporting evidence

The project connects data preparation, supervised and unsupervised analysis, deep learning, explainability, tracking, recommendation and deployment in a local historical-data simulation. Random Forest has provided the lower held-out MAE and is then reused for timing recommendations and the API. Monitoring and input validation has shown the expected behaviour, while group analysis identified weaker conditions and gaps in test coverage.

Before using the system in practice, we would need to collect the data from more time periods and roads, check how well it works in severe weather, and also adjust when monitoring alerts are triggered. People should also review its predictions. To predict actual accidents, we would need real accident data and test whether the model can predict them accurately.

The 7 notebooks in notebooks contain the task workflows. Saved models and preparation artifacts are iorganized n models, MLflow tracking is in mlflow, the recommendation module is in recommendation_system; and the application, manifest and examples are in deployment. Results contains model comparisons, SHAP contributions, validation checks, recommendation outputs and group error tables. The execution output document records screenshots and verification prrof, while the separate responsible and sustainable AI report is in reports. Python logging records task milestones and noteworthy events.

Overall conclusions and reproducibility

The project connects verified historical analysis with repeatable data preparation, model evaluation and a local deployment simulation. Random Forest and the neural network achieved similar held-out performance. Time and weather group analyses identify conditions needing further investigation, especially the limited severe-weather samples.

Part 1 uses fixed congestion thresholds. Part 2 uses cleaned-data percentile categories. Part 3 supervised proxy thresholds are fitted on training data, while its association rules use full historical quartiles. These differences reflect different analytical purposes and are documented in the main README.

The repository is organised as part1_data_analytics, part2_python and part3_machine_learning. Part 2 CLI commands use cli_app/traffic_app.py and figures stores chart outputs. Part 3 monitoring code and the training reference are in monitoring; the deployment application retains its existing monitoring entry point. Models and saved evaluation results were reused during the folder reorganisation.

For reproduction, retain the entire repository, follow the Part 2 script order and use the documented Part 3 notebook working directory. Saved models, preparation settings, hashes, evaluation tables and MLflow records provide traceability. Reports and execution screenshots accompany each part.

Before operational use, collect representative data across more periods and roads, investigate uneven errors, calibrate monitoring thresholds and review predictions with human oversight. Actual accident prediction requires observed accident outcomes and a separately validated model. Energy and emissions were not measured, so no quantified environmental benefit is claimed.